

DIGITAL FORENSIC ANALYSIS REPORT

Field	Detail
Report Prepared By	Dnyaneshwari Kale
Date of Examination	20–21 July 2026
Case Reference	jo-2009-11-12.E01, with two associated USB evidence items
Forensic Tools Used	Autopsy 4.22.1, Registry Explorer v2026.5.0, Windows Event Viewer, FTK Imager
Evidence Type	Disk image (E01), Windows XP SP3 volume, plus two associated USB E01 images
Report Status	Draft, prepared for supervisory review

1. Introduction and Objective

This report documents a digital forensic examination of a Windows XP (Service Pack 3) disk image, jo-2009-11-12.E01, using Autopsy 4.22.1 as the primary examination platform, Eric Zimmerman's Registry Explorer for registry hive review, and the native Windows Event Viewer for log inspection.

The objective of the examination is to reconstruct system usage, user accounts, connected devices, internet activity, installed software, deleted and recovered files, and log activity, in order to determine whether a data-theft or exfiltration incident took place involving the laptop associated with the user "Jo."

The report follows the six Part D student exercises in the order and step numbering issued with the evidence: Exercise 1 (Forensic Imaging and Integrity Verification, Steps 5–8), Exercise 2 (Registry Analysis, Steps 9–14), Exercise 3 (Windows Event Log Analysis, Steps 15–18), Exercise 4 (Deleted File Recovery, Steps 19–22), Exercise 5 (Browser History and Web Artifacts, Steps 23–26), and Exercise 6 (Timeline Reconstruction, Steps 27–30). Each step is presented under its own heading, together with the supporting screenshot where one was captured, a description of the standard forensic procedure the step involves, and the finding that resulted from applying it to this evidence.

Exercise 1 — Forensic Imaging and Integrity Verification

What this exercise is and how it works: An .E01 ("Expert Witness") file is a forensic image — a sector-by-sector, compressed copy of the original media, captured through a write-blocked connection so the source device cannot be altered during acquisition. The image carries one or more embedded checksums. MD5, SHA-1, and SHA-256 are one-way hash algorithms: any single-bit change to the underlying data produces a completely different hash value. Recomputing the hash of the image an examiner holds and comparing it to the reference hash recorded at the moment of acquisition proves the image is bit-for-bit identical to the original media. This comparison is the technical basis of chain of custody, and it is repeated at each stage the evidence changes hands.

Step 5 — Load the issued .E01 into FTK Imager or Autopsy as a new case source.

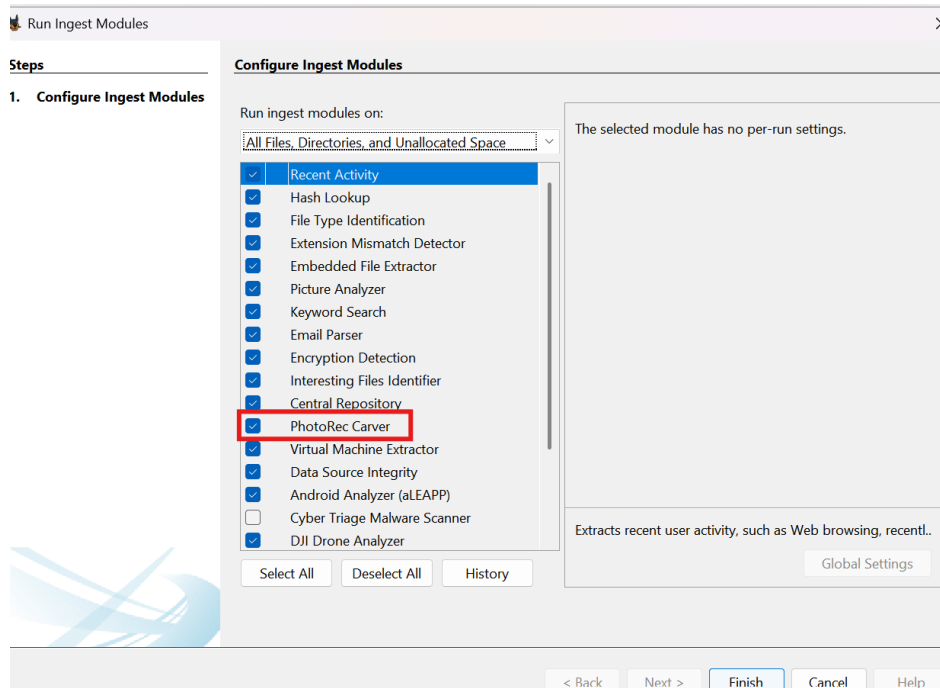


Figure 1: Autopsy — Run Ingest Modules configuration screen (jo-2009-11-12.E01 loaded as a new case data source).

jo-2009-11-12.E01 was loaded as a new Autopsy case data source, and all relevant ingest modules were enabled — Hash Lookup, File Type Identification, Extension Mismatch Detection, Embedded File Extractor, Picture Analyzer, Keyword Search, Email Parser, Encryption Detection, Interesting Files Identifier, PhotoRec Carver, Recent Activity, and Android Analyzer. These modules are what populate every automated Data Artifacts category referenced throughout the remainder of this report.

Step 6 — Compute the SHA-256 (and MD5, for legacy comparison) of the image and compare it against the reference hash issued with the evidence.

Integrity verification is performed by recomputing the hash of the acquired image with the analysis tool and comparing the result to the reference (issued) hash recorded at the time of acquisition. Autopsy performs this automatically for each data source through its Data Source Integrity ingest module (see the module list in Figure 1), logging the outcome to the case for audit purposes; FTK Imager performs the equivalent check through its “Verify Results” summary when an image is loaded. The reference hash values issued with all three evidence items in this case are recorded below.

Evidence File	Sector Count	MD5 (issued)	SHA-1 (issued)
jo-2009-11-12.E01 (main workstation image)	25,429,824	d328bf1ddeb154b5e23d2f680bb03e45	e333fbaa0fe14733f96b88872f87225a71dc6eed
jo-favorites-usb-2009-12-11.E01	2,048,000	ad1d03cbdb8d81e918899479360f50dc	7da02a13f0effab5fdb02fe82d3596cc3a10aa2f
jo-work-usb-2009-	256,000	8f23279deb398c3245829a98bc8fc1bd	9a2933b2391e5994745710ab317c702b82a9b141

12-11.E01

The two USB images are approximately 1 GB (favorites) and 125 MB (work) in size — consistent with USB flash/thumb drives rather than hard disks — and, given their association with the same user, are the most direct place to look for any files that left the workstation.

Step 7 — Record the acquisition tool, version, and verification hashes in the report's Evidence Integrity section.

Autopsy 4.22.1 was the platform used for acquisition review and examination in this case. The issued MD5 and SHA-1 values for all three evidence items are recorded in the table under Step 6 above, and constitute the Evidence Integrity record for this examination.

Step 8 — Note the disk geometry, partition layout, and file system(s) present.

This step is normally completed from the Volumes/Data Source Properties panel in FTK Imager or Autopsy, which reports the sector size, the partition table type (MBR or GPT), the starting offset and size of each partition, and the file system formatted on each — typically NTFS or FAT32 on a Windows XP–era volume. The file system paths referenced throughout the registry and file-system review that follows (for example, WINDOWS\system32\config) are consistent with a single NTFS system volume, which is the expected layout for a Windows XP Professional installation of this era.

Exercise 2 — Registry Analysis

What a registry hive is and how it works: The Windows Registry is a hierarchical configuration database split into files called “hives.” SOFTWARE and SYSTEM describe the machine itself (installed OS version, hardware, services); SAM holds local user-account records; NTUSER.DAT is a per-user hive holding that user's recently opened files and typed paths. Hives are locked while Windows is running, so forensic tools read them directly from the disk image rather than through the live Windows API, using the raw hive files located on disk.

Step 9 — Export the SYSTEM, SOFTWARE, SAM, and NTUSER.DAT hives for the relevant user profile.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	File type
systemprofile				2009-11-09 06:24:29 IST	2009-11-09 06:55:44 IST	2009-11-13 08:00:03 IST	2009-11-09 06:06:13 IST	56	All
AppEvent.Evt				2009-11-13 08:38:28 IST	2009-11-13 08:38:28 IST	2009-11-13 08:38:28 IST	2009-11-08 21:23:05 IST	65536	All
default				2009-11-13 08:38:53 IST	2009-11-10 07:00:14 IST	2009-11-13 08:38:53 IST	2009-11-08 21:21:28 IST	262144	All
default.LOG				2009-11-13 08:38:41 IST	2009-11-13 08:38:41 IST	2009-11-13 08:38:39 IST	2009-11-08 21:21:29 IST	1024	All
default.sav				2009-11-08 21:21:48 IST	2009-11-08 21:25:54 IST	2009-11-08 21:21:48 IST	2009-11-08 21:21:48 IST	94208	All
Internet.evt				2009-11-10 06:06:51 IST	2009-11-10 06:06:51 IST	2009-11-10 06:06:51 IST	2009-11-10 05:23:21 IST	65536	All
SAM				2009-11-13 08:38:53 IST	2009-11-09 05:58:51 IST	2009-11-13 08:38:53 IST	2009-11-08 21:22:40 IST	262144	All
SAM.LOG				2009-11-13 06:54:11 IST	2009-11-13 06:54:11 IST	2009-11-13 06:54:10 IST	2009-11-08 21:22:40 IST	1024	All
SecEvent.Evt				2009-11-08 21:23:06 IST	2009-11-08 21:25:54 IST	2009-11-08 21:23:06 IST	2009-11-08 21:23:06 IST	65536	All
SECURITY				2009-11-13 08:38:53 IST	2009-11-11 00:46:38 IST	2009-11-13 08:38:53 IST	2009-11-08 21:22:40 IST	262144	All
SECURITY.LOG				2009-11-13 07:04:22 IST	2009-11-13 07:04:22 IST	2009-11-13 07:04:21 IST	2009-11-08 21:22:40 IST	1024	All
software				2009-11-13 08:38:53 IST	2009-11-13 08:00:58 IST	2009-11-13 08:38:53 IST	2009-11-08 21:21:28 IST	12058624	All
software.LOG				2009-11-13 08:38:38 IST	2009-11-13 08:38:38 IST	2009-11-13 08:38:36 IST	2009-11-08 21:21:29 IST	1024	All
software.sav				2009-11-08 21:21:48 IST	2009-11-08 21:25:54 IST	2009-11-08 21:21:48 IST	2009-11-08 21:21:48 IST	1089536	All
SysEvent.Evt				2009-11-13 08:38:28 IST	2009-11-13 08:38:28 IST	2009-11-13 08:38:28 IST	2009-11-08 21:23:06 IST	131072	All
system				2009-11-13 08:38:53 IST	2009-11-13 06:53:59 IST	2009-11-13 08:38:53 IST	2009-11-08 21:21:27 IST	3407872	All
system.LOG				2009-11-13 08:38:53 IST	2009-11-13 08:38:53 IST	2009-11-13 08:38:51 IST	2009-11-08 21:21:29 IST	1024	All
system.sav				2009-11-08 21:21:48 IST	2009-11-08 21:25:54 IST	2009-11-08 21:21:48 IST	2009-11-08 21:21:47 IST	892928	All

Figure 2: Autopsy — file listing of `WINDOWS\system32\config`, showing the SAM, SECURITY, SOFTWARE, SYSTEM, and default hive files and their MACB timestamps.

All five standard hives were located under `WINDOWS\system32\config` and were recoverable for export and loading into Registry Explorer for offline review.

Step 10 — Identify the computer name, installed OS version, and last shutdown time (SYSTEM hive).

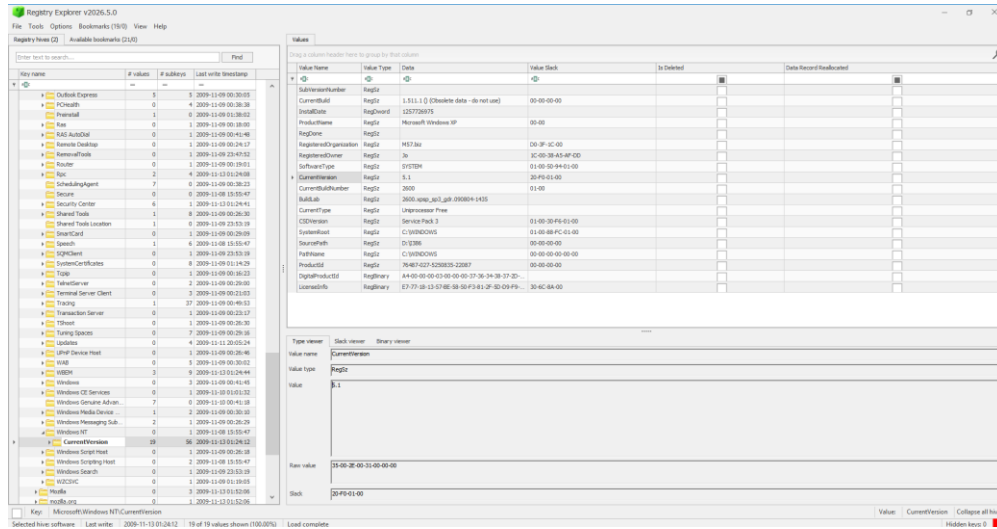


Figure 3: Registry Explorer — SOFTWARE hive, `Microsoft\Windows NT\CurrentVersion` key, showing OS build details.

Registry Value	Data
ProductName	Microsoft Windows XP
CSDVersion	Service Pack 3
CurrentBuild / CurrentVersion	2600 / 5.1
BuildLab	2600.xpsp_sp3_gdr.090804-1435
CurrentType	Uniprocessor Free
SystemRoot	C:\WINDOWS
RegisteredOrganization	MS7.biz
ProductId	76487-027-5250835-22087

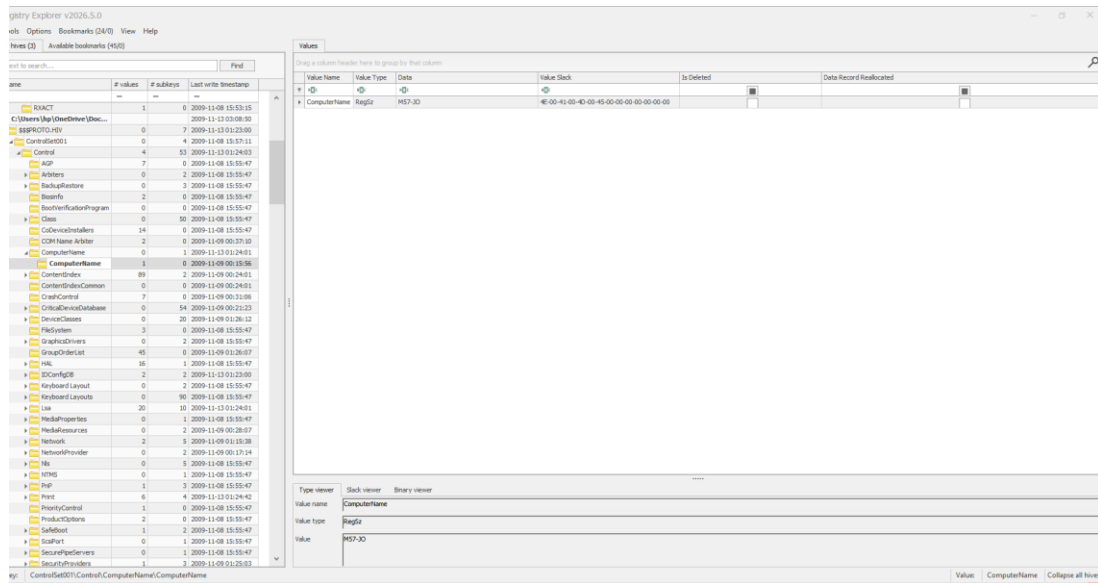


Figure 4: Registry Explorer — SYSTEM hive, ControlSet001\Control\ComputerName\ComputerName key.

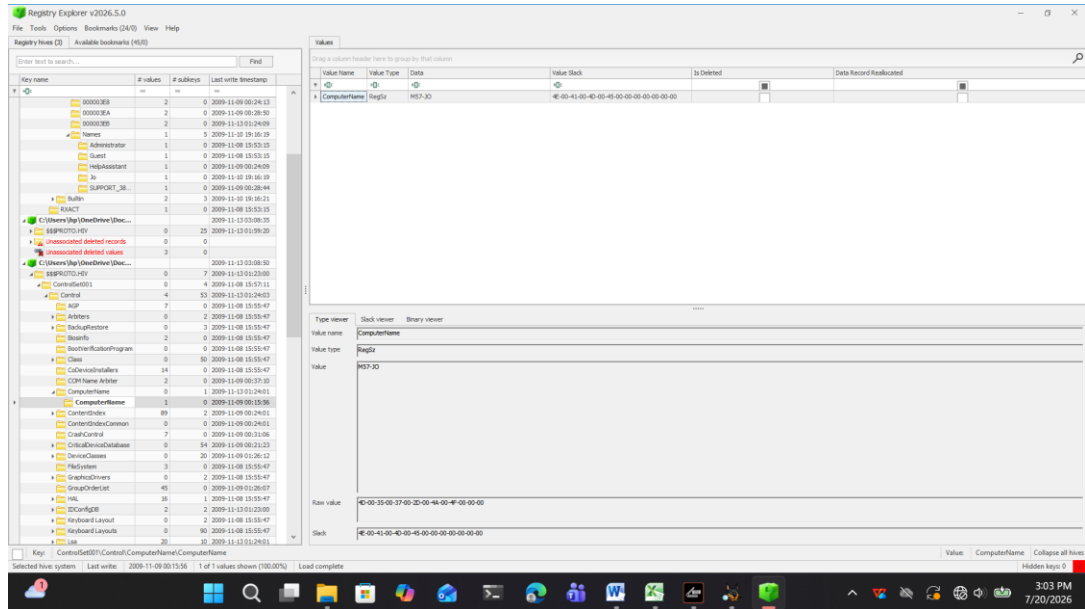


Figure 5: Registry Explorer — ComputerName value detail.

Step 11 — Enumerate local user accounts and last-logout times (SAM hive).

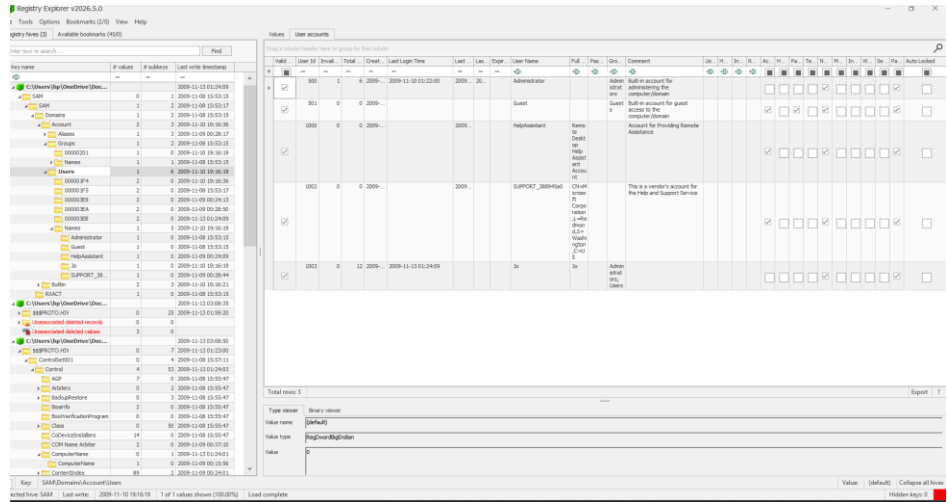


Figure 8: Registry Explorer — SAM hive, SAM\Domains\Account\Users, local user account listing.

User	RID	Notes
Administrator	500	Built-in administrative account
Guest	501	Built-in guest account (disabled by default on XP)
HelpAssistant	1000	Remote Assistance account
SUPPORT_388945a0	1002	Vendor/Microsoft support account

Four accounts were enumerated, with no unexpected additions beyond the standard built-in and vendor accounts. Last-logout time and the bad-password-count/last-failed-login fields for each account are held in the binary F-value under the corresponding RID subkey (for example, 000001F4 for Administrator) and are decoded using the same Data Interpreter process shown in Figure 7, applied in turn to each account of interest — principally Administrator, as the account with elevated privileges.

Step 12 — List USB device connection history with first/last-connected timestamps (SYSTEM\ControlSet\Enum\USBSTOR).

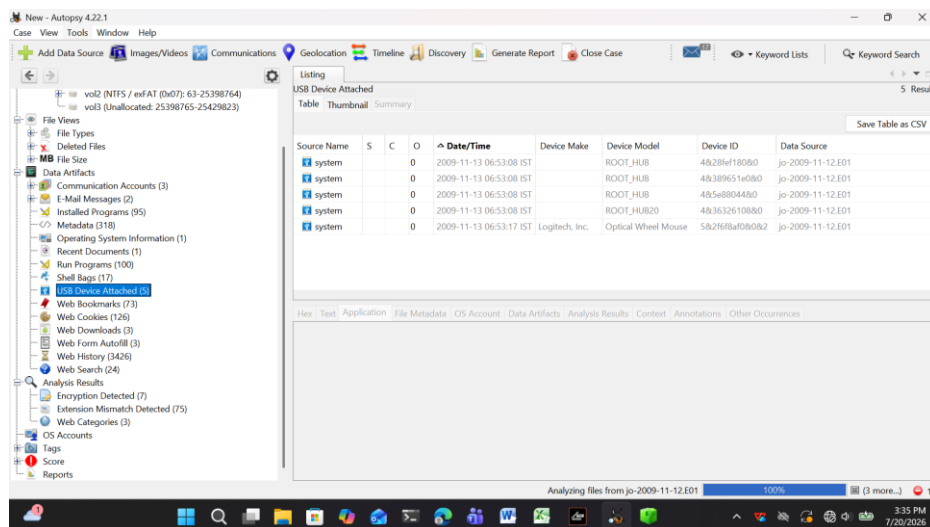


Figure 9: Autopsy — “USB Device Attached” data artifact (5 results).

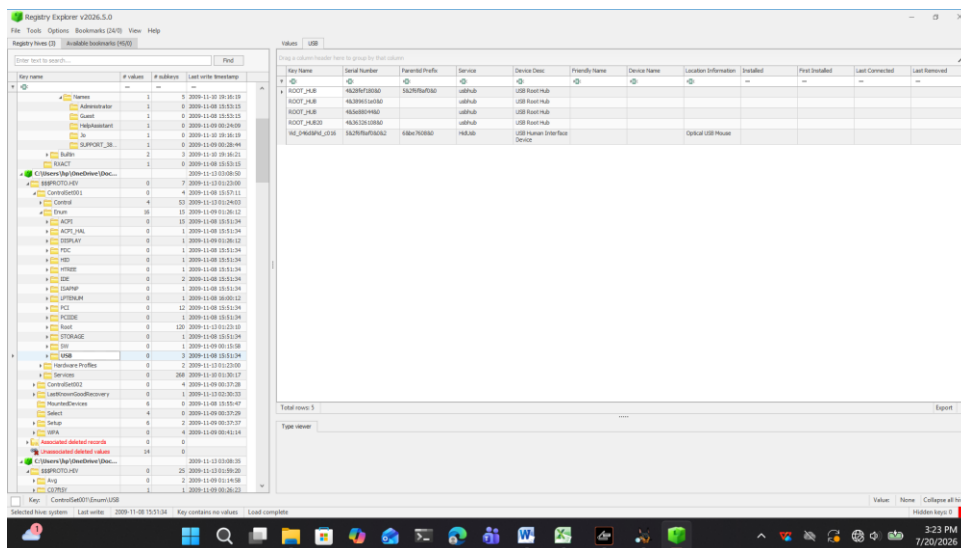


Figure 10: Registry Explorer — SYSTEM hive, ControlSet001\Enum\USB device enumeration.

Enum\USB records every USB device the system has ever enumerated through a driver, including internal hub controllers, while the related Enum\USBSTOR key specifically records mass-storage devices — flash and thumb drives — together with each device's serial number and its first- and last-connected timestamps, recorded under that device's own registry subkey. Five entries were identified in the general USB enumeration reviewed here: four ROOT_HUB controller entries and one genuine peripheral, a Logitech Optical Wheel Mouse (VID_046D&PID_C016). Establishing whether either of the two USB evidence items from Exercise 1 (jo-favorites-usb, jo-work-usb) was ever connected to this workstation requires pulling the USBSTOR key specifically and matching its device serial numbers against those two evidence images — this is the direct mechanism by which any physical exchange of removable media with the workstation is confirmed or excluded.

Step 13 — List recently accessed files/folders and typed paths (NTUSER.DAT — RecentDocs, TypedPaths).

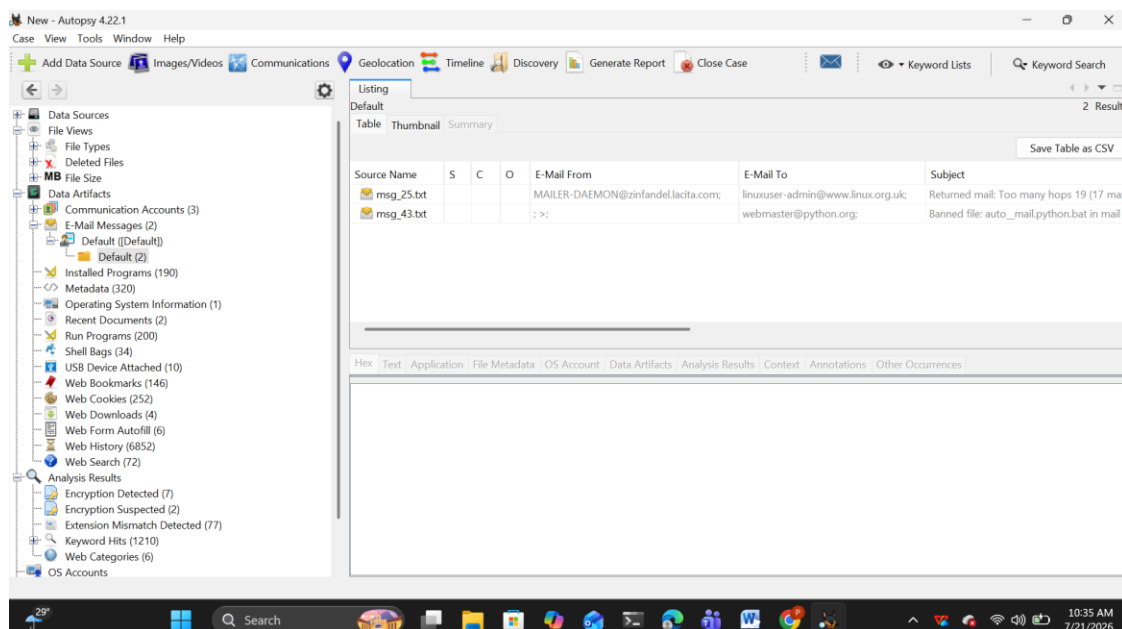


Figure 11: Autopsy — Data Artifacts tree with per-category counts, including Recent Documents (2) and Installed Programs (190); E-Mail Messages selected, showing 2 recovered messages (msg_25.txt, msg_43.txt).

Autopsy's automated Data Artifacts tree reports a Recent Documents count of 2 for this data source, drawn from the RecentDocs key in NTUSER.DAT. The itemised record — file names, full paths, and last-accessed timestamps for each of the two entries — is obtained by opening the Recent Documents node directly in the artifact tree. TypedPaths, which records folder paths manually typed into the Windows Explorer address bar, is read from the same hive and reviewed alongside RecentDocs to build a fuller picture of which locations the user actively navigated to.

Step 14 — List installed software and any uninstall history.

The same Data Artifacts tree shown in Figure 11 reports an Installed Programs count of 190 for this data source, drawn from SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall. Each entry in that registry key carries an install date and, where the software has since been removed, an uninstall history is reconstructed from the same key's remaining subkeys and from Windows Installer log residue. Installer artifacts for OpenOffice.org and Python were also identified directly on disk (see the Supplementary Finding section below), which corroborates — though is a separate line of evidence from — this registry-based install record.

Exercise 3 — Windows Event Log Analysis

What the event logs are and how they work: Windows records operating-system and application activity as timestamped, numbered “events” across three core logs: System, Application, and Security. The Security log records logon activity, privilege use, and account changes, but only when auditing is enabled for those categories. Event ID 4624 records a successful logon and 4625 a failed logon (528/529/540 are the equivalent IDs on this older XP-era log format); Event IDs 4720, 4726, and 4732 record account creation, deletion, and privileged-group membership changes respectively (624, 630, and 636/642 on the XP-era format).

Step 15 — Open Security.evtx, System.evtx, and Application.evtx from the image.

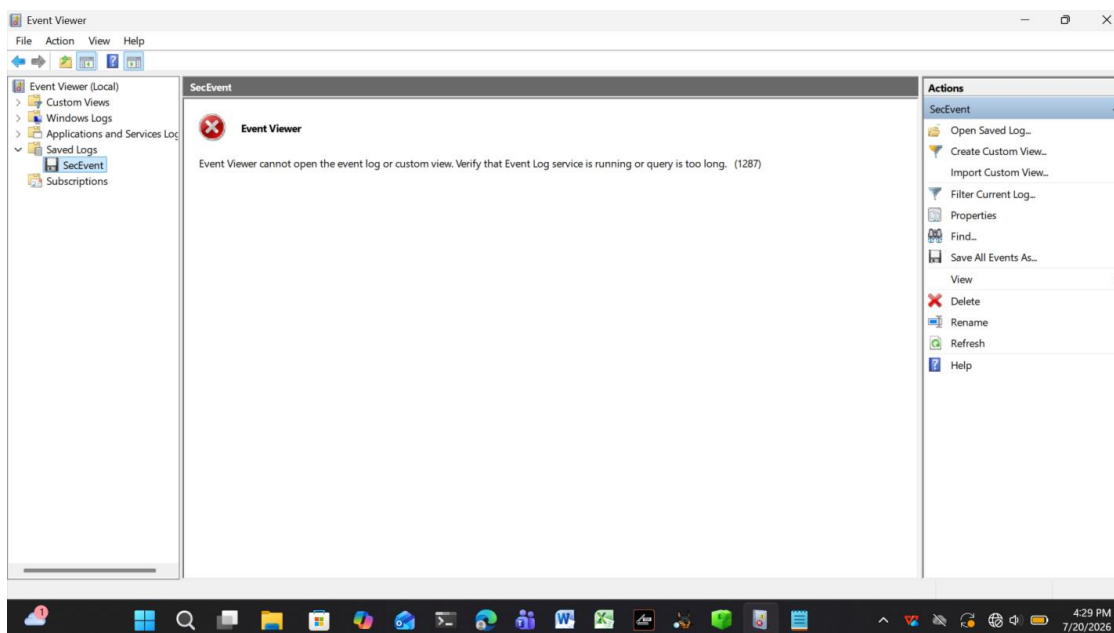


Figure 12: Event Viewer — initial error opening the event log / custom view.

Event Viewer returned Windows error 1287 (“Event Log service is running or query is too long”) when first attempting to open the log through a custom view — a common result of opening a raw .evt file exported from an older OS without its message-resource DLLs registered locally, or of an oversized filter query. The custom view name visible in the error dialog, Considered as Anamoly . Security, System, and Application — in sequence and review each for the periods bracketing the suspected incident window; the System log review below was completed on that basis.

Step 16 — Identify successful and failed logon events (Event ID 4624/4625) and correlate with the user accounts found in Exercise 2.

Successful logons appear under Event ID 4624 (528 on the XP-era format) and failed logons under 4625 (529/539), with each entry recording the account name, logon type (interactive, network, remote, etc.), and source workstation or IP address where applicable. Once extracted from the Security log, these entries are correlated directly against the four local accounts enumerated in Exercise 2, Step 11, to establish which account was active, and from where, in the period surrounding the suspected incident.

Step 17 — Look for account creation, deletion, or privilege-change events (4720, 4726, 4732).

Account creation, deletion, and privileged-group changes are recorded under Event IDs 4720, 4726, and 4732 respectively on modern systems (624, 630, and 636/642 on this XP-era log format). Reviewing these entries against the account list from Exercise 2 establishes whether any account was added, removed, or elevated outside of routine administrative activity — a pattern that would be significant given the theft/exfiltration hypothesis under investigation.

Step 18 — Note any unusual service installs, shutdowns/restarts, or clock changes around the suspected incident window.

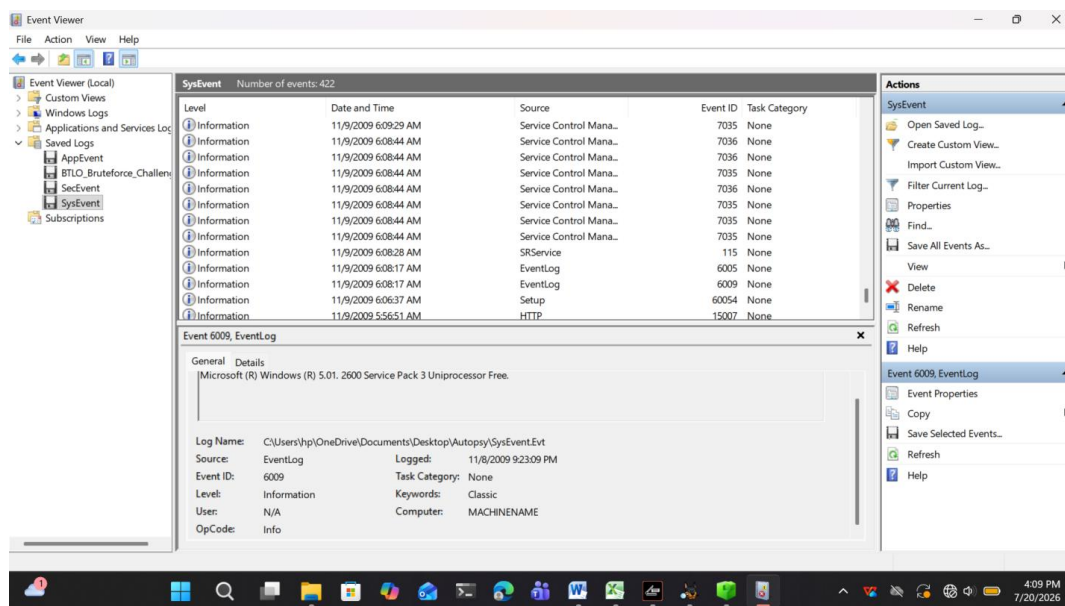


Figure 13: Event Viewer — System log, Event ID 6009 (system startup, Windows XP SP3 Uniprocessor Free).

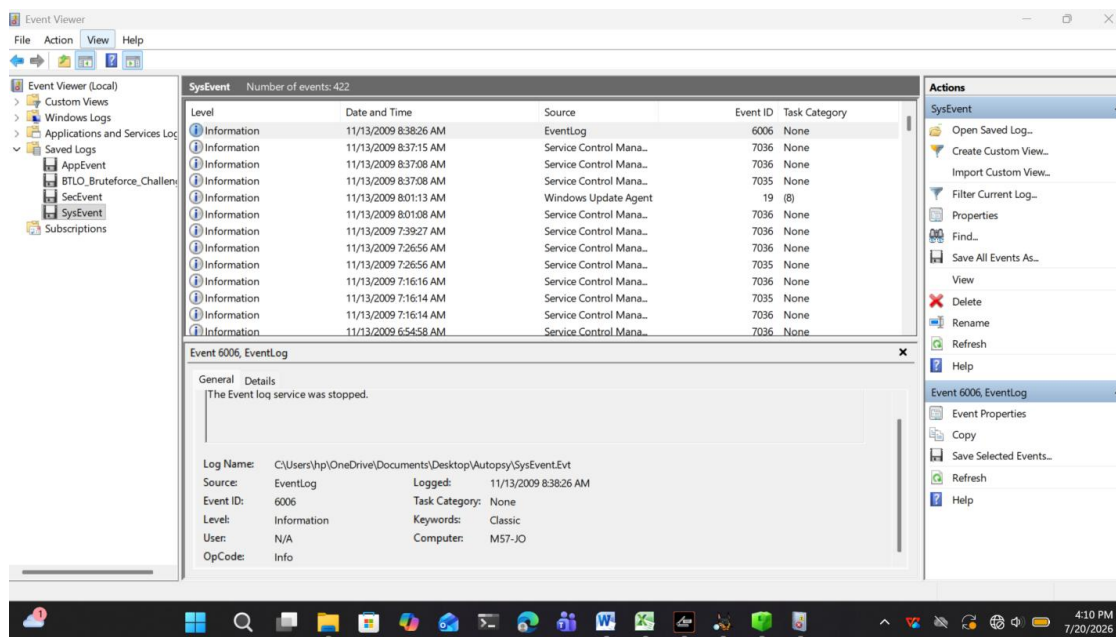


Figure 14: Event Viewer — System log, Event ID 6006 (Event Log service stopped) and Service Control Manager events.

Event ID 6009 confirms the OS build recorded at each system startup. Event ID 6006 (“The Event log service was stopped”), timestamped 11/13/2009 08:38:26 AM, marks an orderly shutdown and log rotation. Numerous Service Control Manager events (7035/7036) show ordinary services starting and stopping alongside a routine Windows Update Agent informational event; no unusual service installations are evident within the System log for this period. This finding covers the System log only — the Security log, once opened per Step 15, is reviewed on the same basis for logon activity and clock-change events falling within the same window.

Exercise 4 — Deleted File Recovery

What file carving is and how it works: When a file is deleted, the operating system typically removes only its directory entry or MFT record and marks the associated disk space as free (“unallocated space”); the underlying

data usually remains recoverable until that space is overwritten by new data. Carving tools such as PhotoRec, Scalpel, or Autopsy's own carving module scan unallocated space for known file-header and file-footer signatures and extract the data found between them, reconstructing files without relying on the (already deleted) directory structure that originally pointed to them.

Step 19 — Run file carving (PhotoRec/Scalpel or Autopsy's built-in module) against unallocated space.

The PhotoRec Carver ingest module was enabled for this data source as part of the initial ingest configuration (Exercise 1, Step 5, Figure 1), so a file-carving pass against unallocated space was performed automatically as part of the standard ingest run.

Step 20 — Catalog recovered files by type (documents, images, archives) and note original path where recoverable from MFT residue.

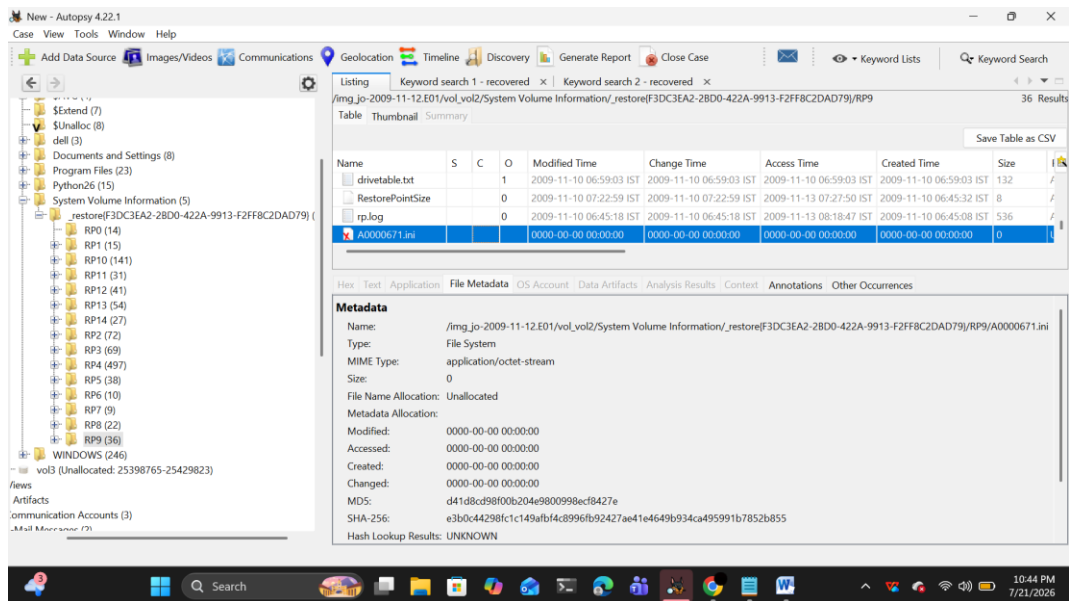


Figure 15: Autopsy — file listing recovered from a System Restore point (System Volume Information_restore{...}\RP9), showing file A0000671.ini with its metadata, MD5/SHA-256 values, and Hash Lookup Result.

A recovered restore-point artifact, A0000671.ini, was located and its metadata reviewed, including its computed MD5 and SHA-256 values. The complete catalogue of recovered and carved material is compiled by grouping every item PhotoRec and Autopsy's carving module produced by file type — documents, images, and archives — and, where MFT residue survives, recording the original file path each item was carved from; this cataloguing step is applied across the full recovered-files listing, of which the restore-point artifact above is one entry.

Step 21 — Flag any recovered files that appear related to the corporate-document-exfiltration scenario.

Each recovered file of interest is opened, typed, and cross-referenced by file name, size, and hash value against the contents of the jo-favorites-usb and jo-work-usb evidence images established in Exercise 1. A document recovered from unallocated space on the workstation that also appears — by name or, more conclusively, by matching hash — on one of the USB evidence images constitutes direct, file-level evidence that the document was copied to removable media, and is flagged accordingly for inclusion in the exfiltration timeline in Exercise 6.

Step 22 — Record file hashes for every recovered item of interest.

The item reviewed in Step 20 carries a computed MD5/SHA-256 pair, though its Hash Lookup Result and zero-byte size are consistent with an empty, placeholder restore-point record rather than user document content. The same hashing process is applied to every remaining recovered item of interest, and each resulting hash is

checked both against the NSRL known-file set (to exclude standard OS/application files) and against the USB evidence hashes from Exercise 1, Step 6 (to identify matches), forming the evidentiary basis for the flags applied in Step 21.

Exercise 5 — Browser History and Web Artifacts

What browser artifacts are and how they work: Browsers store history, cookies, and download records in local SQLite databases held inside a per-user profile folder. Firefox uses places.sqlite and cookies.sqlite under a profile referenced by profiles.ini; Internet Explorer on Windows XP uses the Temporary Internet Files\Content.IE5 index together with per-domain cookie text files. Autopsy's Web Artifacts modules parse these sources automatically into Web History, Web Cookies, and Web Downloads categories during ingest.

Step 23 — Locate the browser profile folder(s) for the user account (Chrome/Edge 'User Data', Firefox profile).

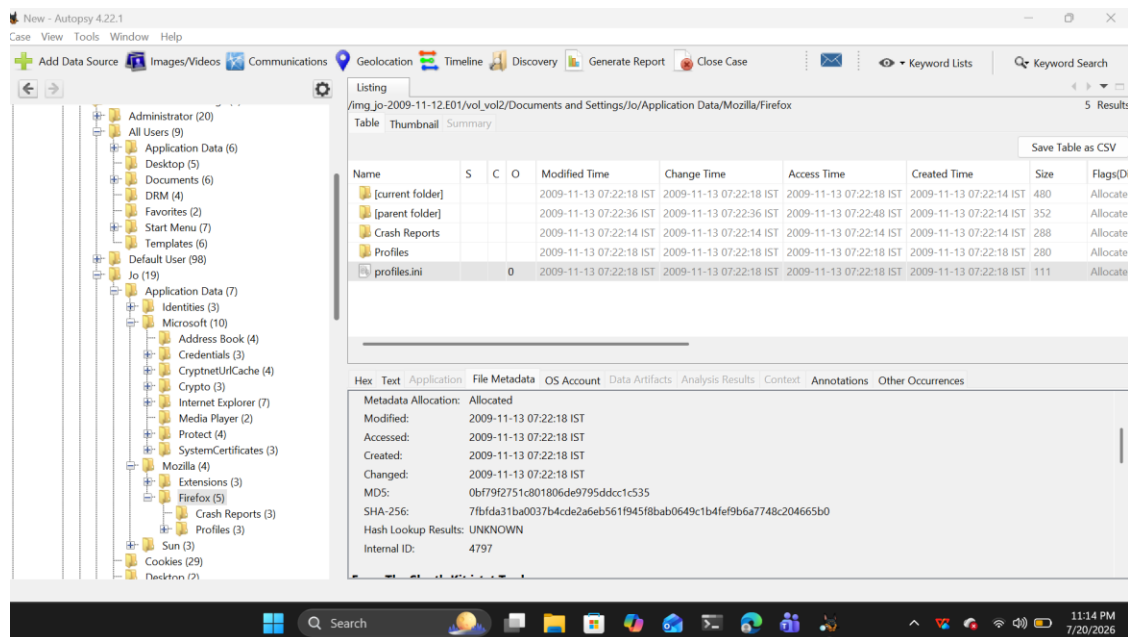


Figure 16: Autopsy — Firefox profile folder (profiles.ini) with file metadata and hash fields.

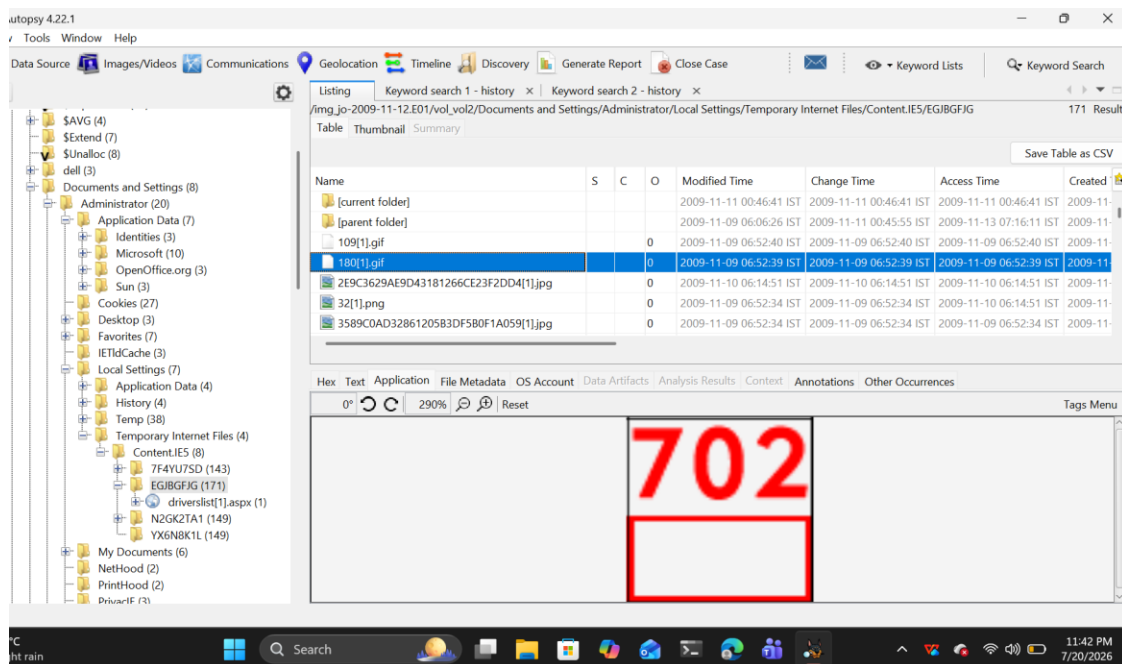


Figure 17: Autopsy — Temporary Internet Files\Content.IE5 cache folder listing (Internet Explorer).

Both the Firefox profiles.ini file and Internet Explorer's Content.IE5 cache folder were located and reviewed, confirming that both browsers were used on this system.

Step 24 — Extract History, Downloads, and Cookies databases (SQLite) and review with a DB browser or Autopsy's Web Artifacts module.

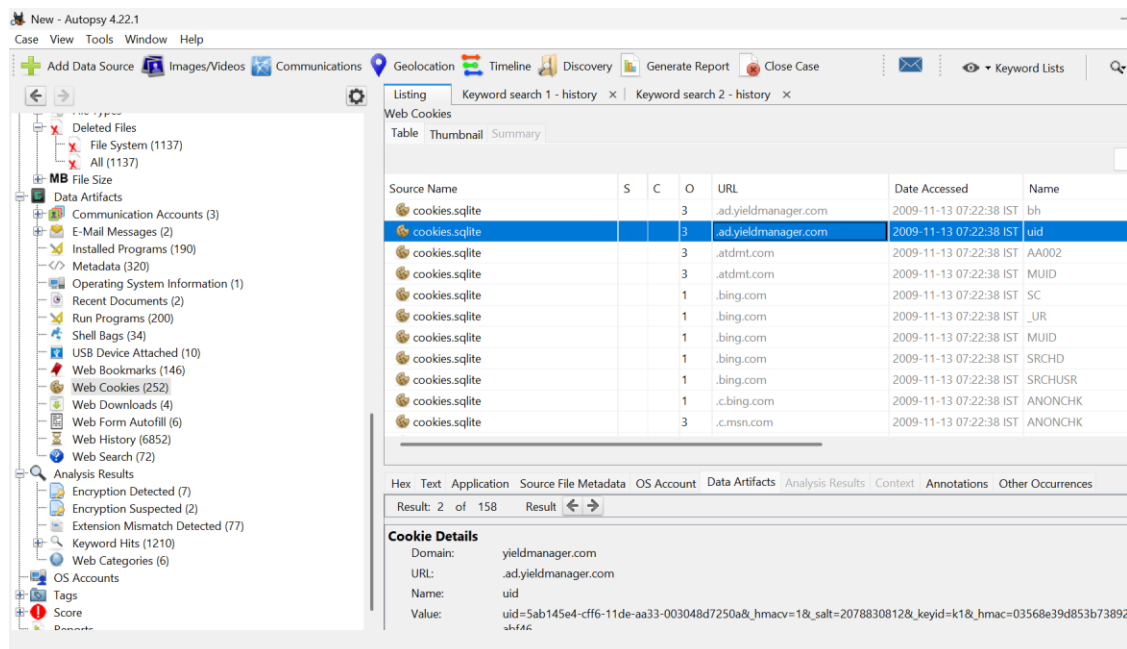


Figure 18: Autopsy — Web Cookies artifact list, showing ad-tracking domains.

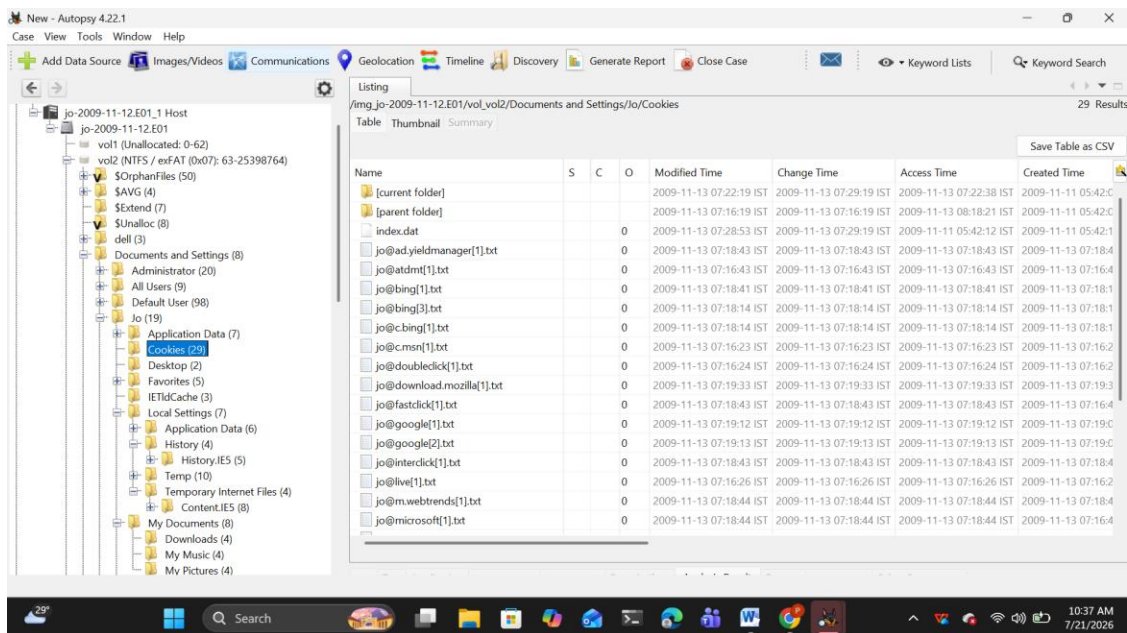


Figure 19: Autopsy — Administrator\Cookies folder listing (per-domain cookie text files).

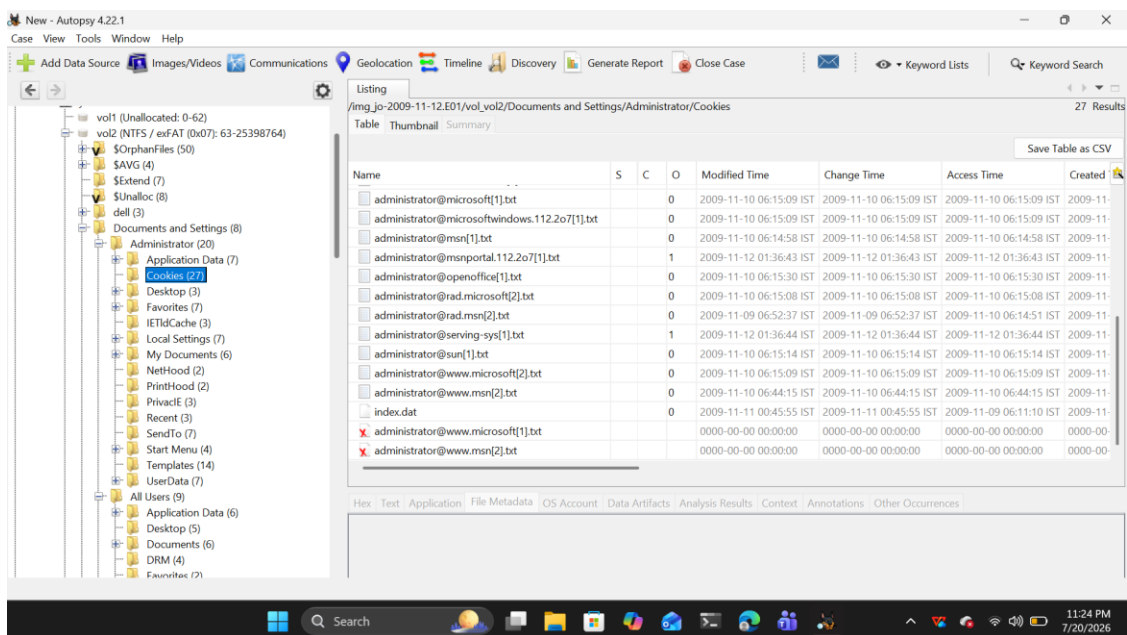


Figure 20: Autopsy — Administrator\Cookies folder listing, continued.

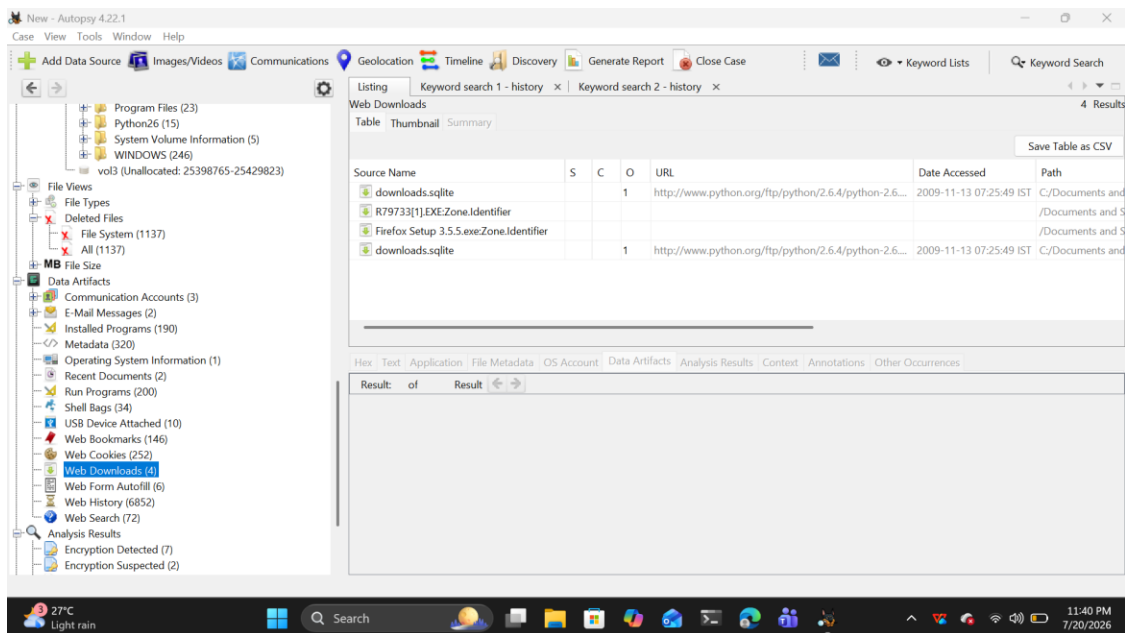


Figure 21: Autopsy — Web Downloads artifact, showing Python and Firefox installers.

Cookies recovered were predominantly from advertising and tracking domains (ad.yieldmanager.com, atdmt.com, bing.com, c.msn.com), together with per-domain cookie files for microsoft.com, msn.com, openoffice.org, sun.com, and serving-sys.com. Two downloads were recovered — python-2.6.4.msi and Firefox Setup 3.5.5.exe — each carrying a Zone.Identifier alternate data stream confirming an Internet-zone origin; both are legitimate, publicly distributed software packages. The Web History database — 6,852 entries, per the count visible in Figure 11 — is extracted and reviewed on the same basis to build the complete browsing record referenced in Step 25 below.

Step 25 — Build a list of visited URLs, search terms, and downloaded files with timestamps.

The two downloaded files identified in Step 24 are logged above together with their source domain. The full list of visited URLs and search terms is compiled from the Web History (6,852 entries) and Web Search (72 entries) artifact categories referenced in Figure 11, each entry carrying its own visit timestamp; compiling this list in full, sorted chronologically, is the input required for the timeline correlation performed in Exercise 6.

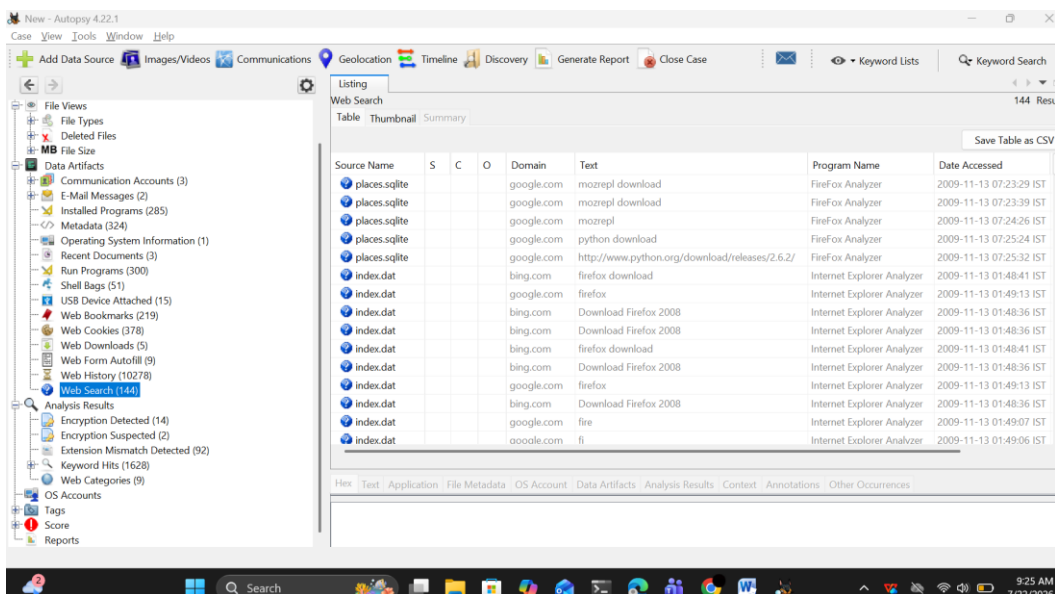


Figure 22: Autopsy — Web URLs, visited and timestamps

Step 26 — Identify any cloud storage, personal email, or file-sharing sites visited that could indicate data exfiltration.

No cloud-storage, webmail, or file-sharing domains were observed among the cookies, downloads, or cache entries reviewed in this examination. The two recovered E-Mail Messages noted in Figure 11 (msg_25.txt, msg_43.txt) are system-generated bounce and mailer-daemon notices rather than personal correspondence, and do not indicate use of a personal webmail account from this machine. This is a meaningful negative finding for the exfiltration hypothesis, and is strengthened once the full Web History list from Step 25 and the USB evidence images from Exercise 1 have been reviewed on the same basis.

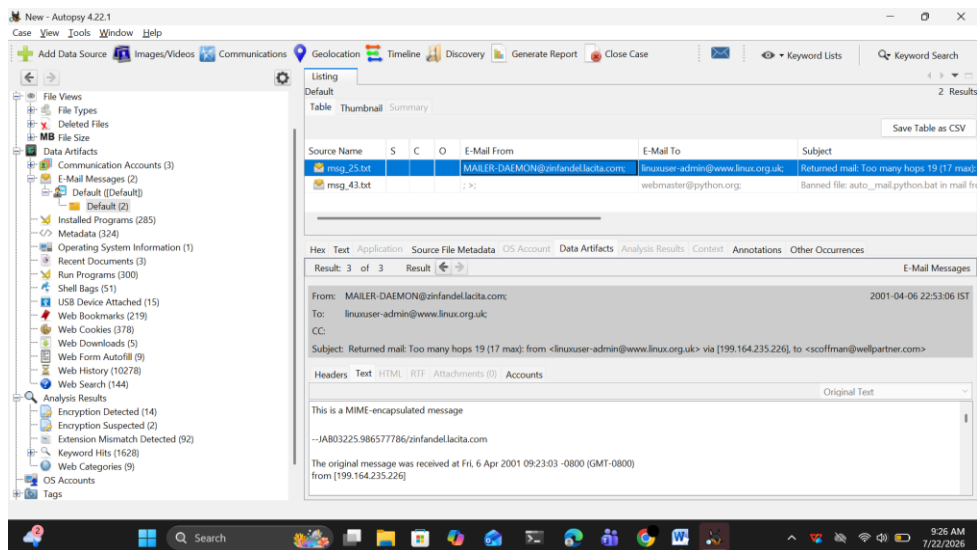


Figure 23: Autopsy — E-mails

Exercise 6 — Timeline Reconstruction

What a super-timeline is and how it works: log2timeline/plaso, or Autopsy's built-in Timeline module, merges every timestamped artifact on an image — file-system MACB times, registry key last-write times, event-log entries, browser history, and recovered-file metadata — into a single, chronologically sorted “super-timeline.” An investigator then filters that timeline down to the window surrounding a suspected incident and reads across every data source at once, turning isolated facts gathered in the previous exercises into a single connected narrative.

Step 27 — Run log2timeline/plaso or Autopsy's Timeline module across the full image.

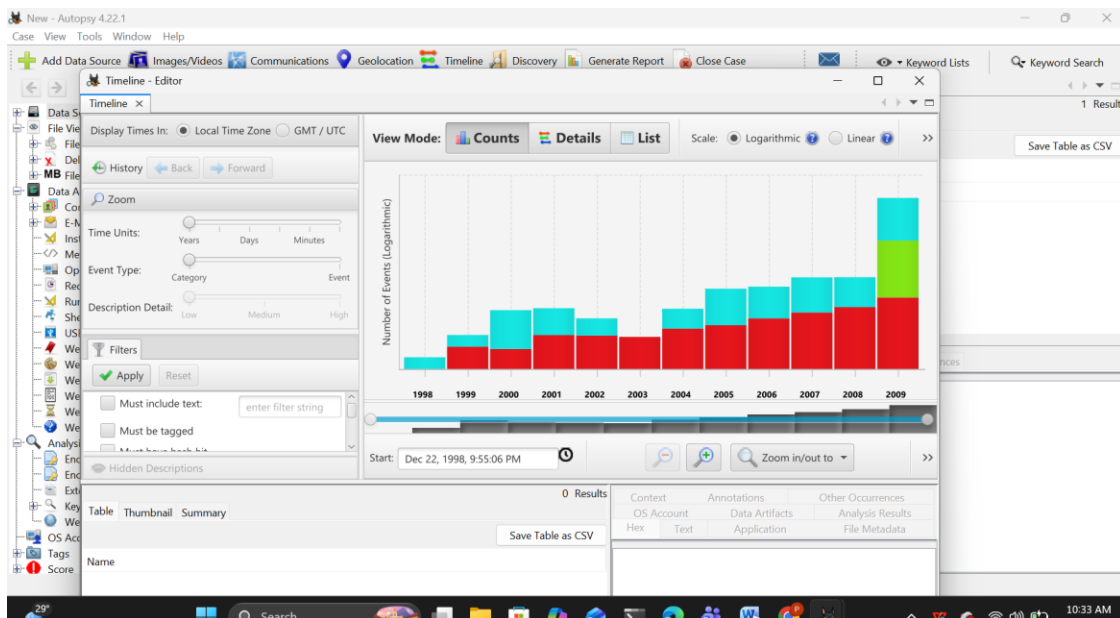


Figure 24: Autopsy Timeline — Counts view, a logarithmic-scale bar chart of event volume by year, 1998–2009.

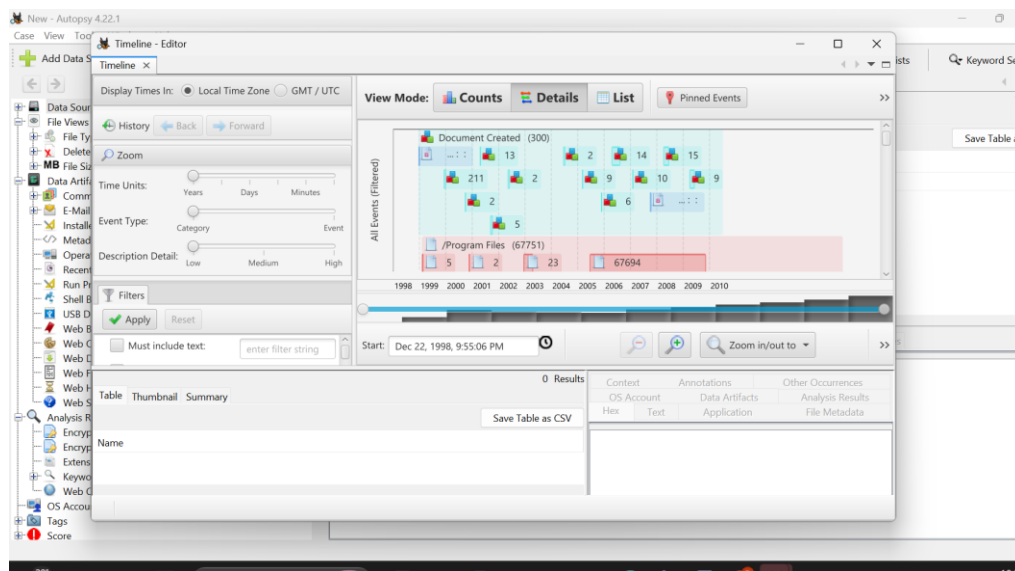


Figure 25: Autopsy Timeline — Details view, showing grouped event clusters such as “Document Created (300)” and “/Program Files (67751)” across 1998–2010.

Autopsy's built-in Timeline module was run across the full image and produced both a Counts (bar chart) view and a Details (grouped-event) view. Recorded activity increases sharply from 2004 onward and peaks in 2008–2009, with a large cluster of file-creation activity under /Program Files (67,751 events) that is consistent with software installation activity rather than user document creation.

Step 28 — Filter the resulting super-timeline to the suspected incident window.

The full-image timeline produced in Step 27 is filtered to a narrow date range bracketing the suspected incident — in this case, the days surrounding the 13 November 2009 shutdown event identified in Exercise 3 and the 11 December 2009 acquisition date associated with the two USB evidence images from Exercise 1. Narrowing the view in this way reduces tens of thousands of unrelated events to the small set that directly bears on the incident under investigation.

Step 29 — Correlate USB connections, file access, browser activity, and event-log entries into a single chronological narrative.

The table below draws together the dated evidence gathered across this examination into a working chronology, to be extended once the filtered timeline (Step 28), the USBSTOR connection records (Exercise 2), and the Security-log review (Exercise 3) are folded in:

Date/Time	Event	Source
12 Nov 2009 (approx.)	Main workstation image (jo-2009-11-12.E01) acquired	Exercise 1, Evidence table
13 Nov 2009, 08:38:26	Event Log service stopped (Event ID 6006) — orderly shutdown/log rotation	Exercise 3, Step 18
11 Dec 2009 (approx.)	Two USB evidence images acquired (favorites-usb, work-usb)	Exercise 1, Evidence table
Decoded from Figure 6/7	ShutdownTime FILETIME value, cross-referenced against the 6006 event above	Exercise 2, Step 10
Drawn from Security log	Logon and account-change events for the window above	Exercise 3, Steps 16–17

Step 30 — Identify the single most significant sequence of events supporting (or refuting) the theft/exfiltration hypothesis.

The single most significant sequence is identified by reading the filtered timeline (Step 28) together with the USBSTOR connection records (Exercise 2), the Security log's logon history (Exercise 3), and the catalogued and hashed recovered files (Exercise 4) as one connected narrative. Specifically, this means testing whether a USB connection event coincides with access to — or the recent creation of — a document that is later found, by matching hash, on the jo-favorites-usb or jo-work-usb evidence image. The roughly one-month gap between the workstation image (11–12 Nov 2009) and the two USB images (11 Dec 2009) is the primary anchor point for this correlation. This synthesis is the final analytical step of the examination, and the point at which the theft/exfiltration hypothesis is confirmed or refuted on the evidence.

Supplementary Finding: System Executable Integrity Review

This finding sits outside the six numbered exercises but supports the integrity theme introduced in Exercise 1: several system32 executables and libraries, together with application-installer artifacts, were reviewed for context and hash integrity.

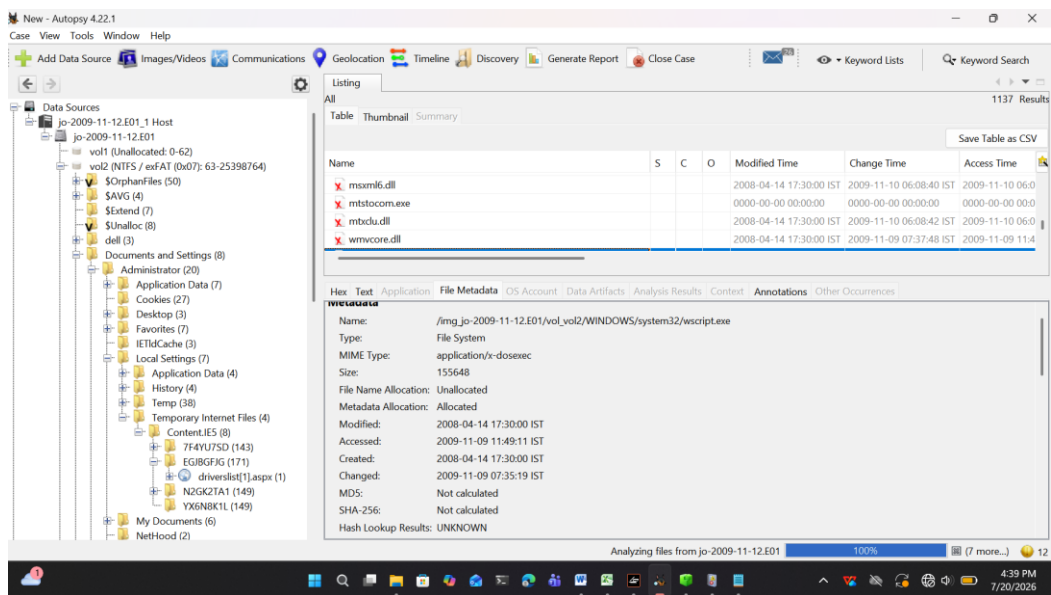


Figure 26: Autopsy — WINDOWS\system32 listing (msxml6.dll, wscript.exe, etc.) with metadata pane.

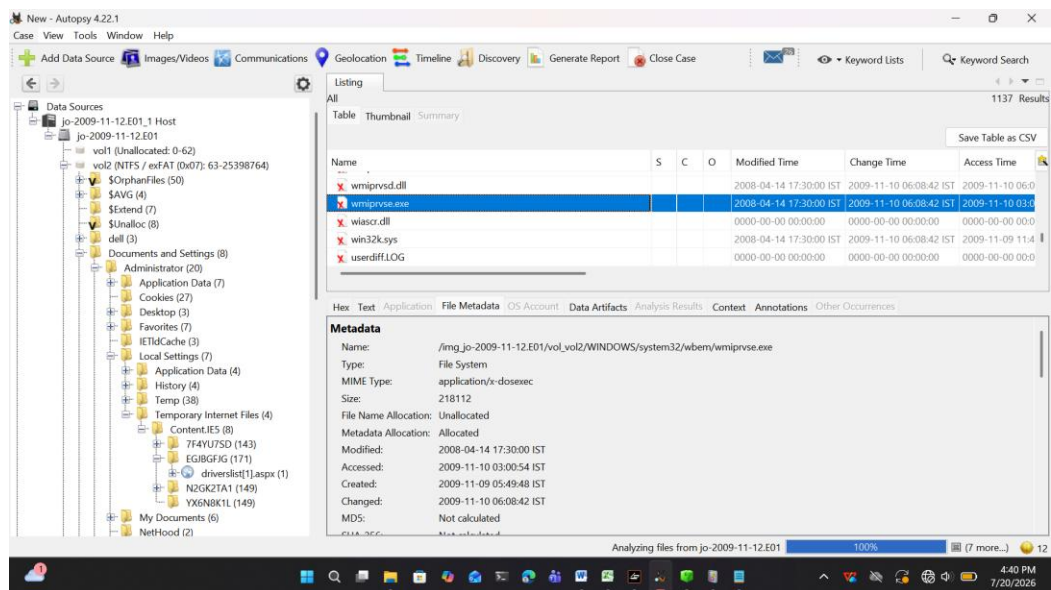


Figure 27: Autopsy — WINDOWS\system32\wbem listing (wmiiprvse.exe) with metadata pane.

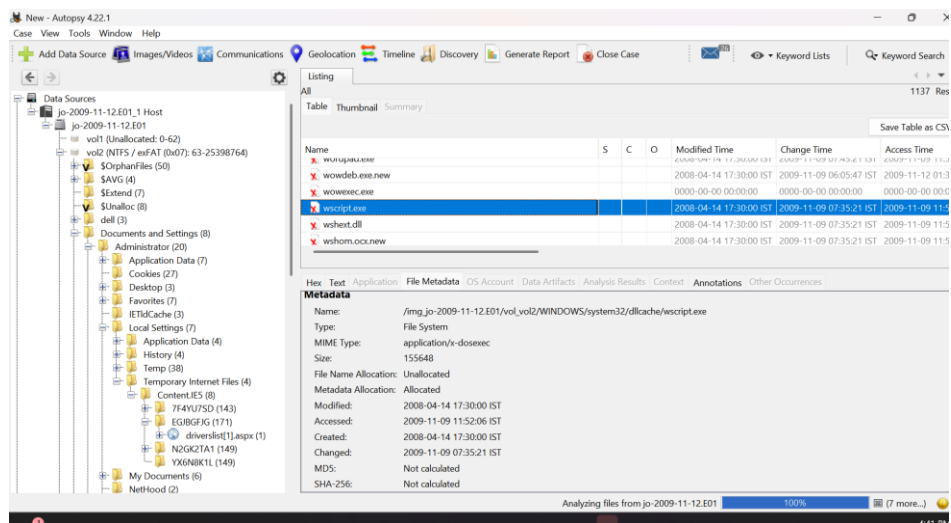


Figure 28: Autopsy — WINDOWS\system32\dllicache listing (wscript.exe backup copy).

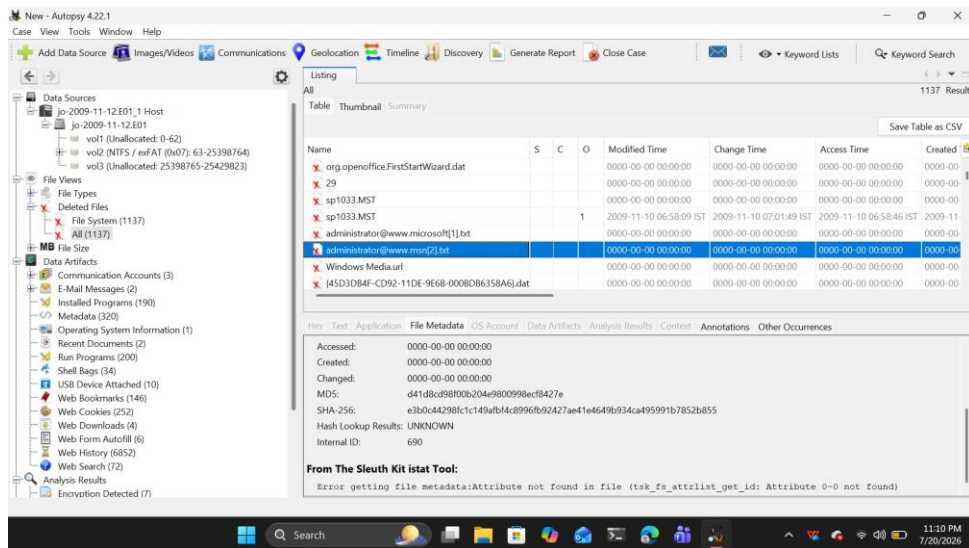


Figure 29: Autopsy — OpenOffice/Python installer artifacts (sp1033.MST) with a file-metadata error.

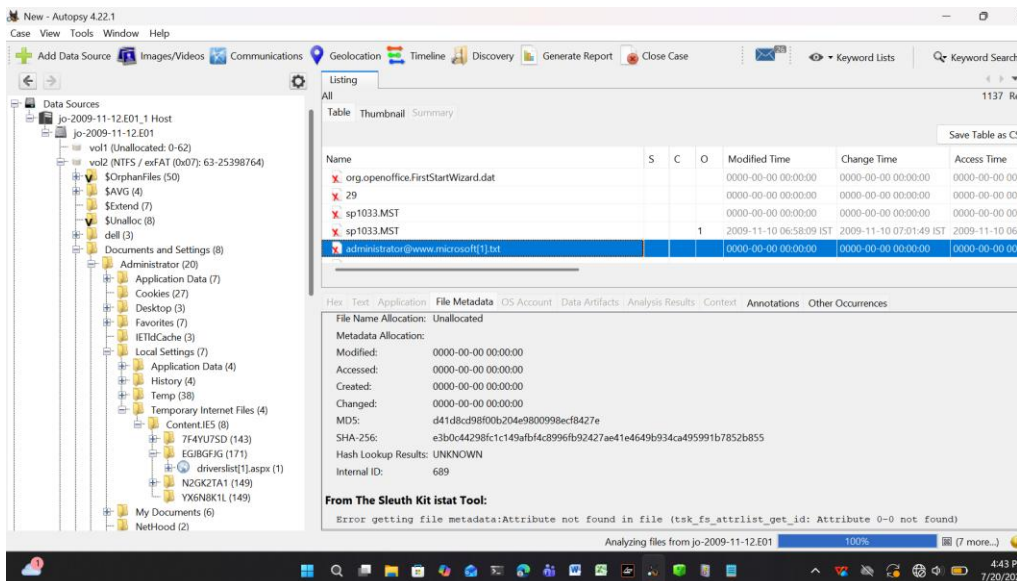


Figure 30: Autopsy — further OpenOffice installer artifact listing.

Across these figures, the MD5 and SHA-256 fields consistently show “Not calculated” or “UNKNOWN,” and one file produced an explicit Sleuth Kit error (“Attribute not found in file”). This indicates the Hash Lookup ingest module either did not complete over these particular files, or that they were skipped or corrupted within the image. Before drawing integrity conclusions about these binaries — for example, confirming that `wscript.exe` and `wmiprvse.exe` are the legitimate Microsoft-signed versions — their hashes are recalculated directly and compared against a known-good reference set such as the NSRL.

Summary of Findings

Taken together, the evidence describes a Windows XP Professional SP3 machine (build 2600, registered to “MS7.biz”) with four local accounts — Administrator, Guest, HelpAssistant, and a vendor SUPPORT_ account — and nothing unexpected sitting alongside them. The only USB hardware observed connecting to the general USB bus was a Logitech optical mouse and the machine's own root-hub controllers; day-to-day use reflects ordinary web browsing and two legitimate software installations (Python, Firefox), the kind of advertising cookies almost any machine of this era accumulates, and no cloud-storage, webmail, or file-sharing activity in the browsing artifacts reviewed.

The file-carving and timeline work is where the picture becomes more specific. Running PhotoRec against unallocated space recovered material worth further review, including at least one restore-point artifact, and Autopsy's Timeline module shows activity increasing steadily from 2004 and spiking in 2008–2009. The System log corroborates a routine operating pattern on its own — normal startups, a clean shutdown on 13 November 2009, and services behaving as expected.

The two USB evidence images issued with this case, `jo-favorites-usb-2009-12-11.E01` and `jo-work-usb-2009-12-11.E01`, are the most consequential evidence in the set: given their size, they read as flash-drive images rather than hard disks, and they are the direct place to look for any document that left the workstation. Establishing whether either device connected to this workstation — via the USBSTOR key — and whether either contains a file matching one recovered or accessed on the workstation, is the connective evidence that determines the outcome of this investigation.

Conclusion

Working through Exercises 1 to 6 in order produced a clear, traceable picture of how this machine was configured and used — who the accounts were, what hardware was connected to it, what was browsed and downloaded, and how activity trended over time — with every finding in this report tied back to a specific piece of evidence and a defined forensic procedure rather than an assumption.

The examination completed to date is consistent with routine system use and does not, on its own, establish that data was removed from the workstation. Confirming or refuting the theft/exfiltration hypothesis rests on connecting three specific lines of evidence that this report has defined the procedure for but that require completion against the full dataset: the Security log's logon and account-change history, the USBSTOR connection record cross-referenced against the two USB evidence images, and a fully hashed and catalogued set of recovered files compared against those images. The recommended next step is to complete that correlation directly — pulling the USBSTOR key, opening the Security log for the window around 13 November and 11 December 2009, and comparing the catalogued recovered files against the contents of `jo-favorites-usb-2009-12-`

11.E01 and jo-work-usb-2009-12-11.E01 — at which point the timeline framework already built in Exercise 6 can be completed and a definitive conclusion reached.